

Section 1.5

Solution Sets of Linear Systems

Homogeneous Linear Systems

A system of linear equations is said to be homogeneous if it can be written in the form $A\mathbf{x} = \mathbf{0}$ where A is an $m \times n$ matrix.

$A\mathbf{x} = \mathbf{0}$ always has the trivial solution $\mathbf{x} = \mathbf{0}$.

$A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the equation has at least one free variable.

Example 1

Determine if the homogeneous system has a nontrivial solution. If so, describe the solution set.

$$\begin{array}{rcl} x_1 + 2x_2 - 3x_3 & = & 0 \\ 2x_1 + x_2 - 3x_3 & = & 0 \\ -x_1 + x_2 & = & 0 \end{array}$$

Parametric Vector Form

Whenever a solution set is described explicitly using a sum of vectors, we say the solution is in parametric vector form.

Example 2

Describe the solutions of the following system in parametric vector form.

$$\begin{array}{rcl} x_1 + 2x_2 - 3x_3 & = & 5 \\ 2x_1 + x_2 - 3x_3 & = & 13 \\ -x_1 + x_2 & = & -8 \end{array}$$

Then, give a geometric description of the solution set.

Theorem

Suppose the solution set of $A\mathbf{x} = \mathbf{b}$ is consistent, and let $\mathbf{p}$ be a solution.

Then, the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form $\mathbf{x} = \mathbf{p} + \mathbf{v}_h$ where $\mathbf{v}_h$ is any solution of the associated homogeneous equation $A\mathbf{x} = \mathbf{0}$.